

E-COMMERCE SALES ANALYTICS DASHBOARD

Interactive Data Analytics & Business Intelligence Project

Component	Technology / Detail
Data Analysis	Python + Pandas
Visualization	Matplotlib
Dashboard	Streamlit
Development	Google Colab
Dataset	1,000 e-commerce orders

An end-to-end analytics project transforming transactional sales data into interactive KPIs, visual trends and actionable business insights.

1. Executive Summary

This project develops an interactive E-Commerce Sales Analytics Dashboard that converts transactional records into business-oriented insights. The dashboard presents key performance indicators, revenue trends, category and regional comparisons, product performance and customer-level analysis.

Metric	Project Result
Total Revenue	■35,572,500.00
Total Orders	1,000
Total Customers	198
Total Quantity Sold	3,051
Average Order Value	■35,572.50

2. Project Objectives

- Analyse sales performance from transactional e-commerce data.
- Create clear KPIs for revenue, orders, customers, quantity and average order value.
- Identify monthly revenue patterns.
- Compare revenue across categories, regions, products and payment methods.
- Provide interactive filtering through a Streamlit dashboard.
- Automatically generate business insights from the filtered dataset.

3. Technology Stack

Technology	Role
Python	Core programming and analytics
Pandas	Data loading, transformation and aggregation
Matplotlib	Monthly revenue visualization
Streamlit	Interactive dashboard and filters
Google Colab	Development environment
Cloudflare Tunnel	Temporary public dashboard demonstration

4. Key Business Results

Area	Best Performer	Revenue / Result
Category	Electronics	■18,174,975.00
Product	Monitor	■5,397,400.00
Region	South	■10,198,825.00
Payment Method	Debit Card	Leading payment method
Month	2025-01	■4,434,725.00

5. Analytical Method

- Loaded the e-commerce transaction CSV into Pandas.
- Converted Order_Date into a datetime field for analysis.
- Calculated KPIs from the filtered transaction dataset.
- Grouped revenue by Month, Category, Region, Product and Payment_Method.
- Used Streamlit controls to dynamically filter the analysis.
- Displayed automated best-performer insights based on the filtered results.

6. Dashboard Screenshots & Visual Evidence

The following screenshots document the completed Streamlit dashboard. Each visual is presented separately with a descriptive figure caption so the report can also serve as portfolio evidence.

6.1 Dashboard Overview & KPIs

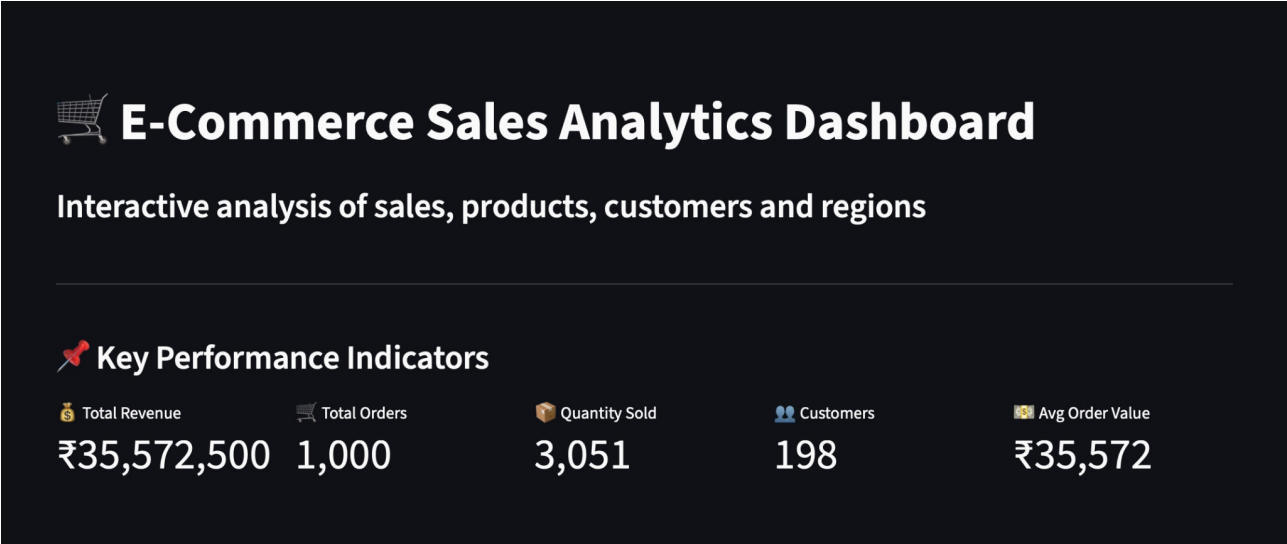


Figure 1. Dashboard overview and Key Performance Indicators (KPIs).

6.2 Monthly Revenue Trend

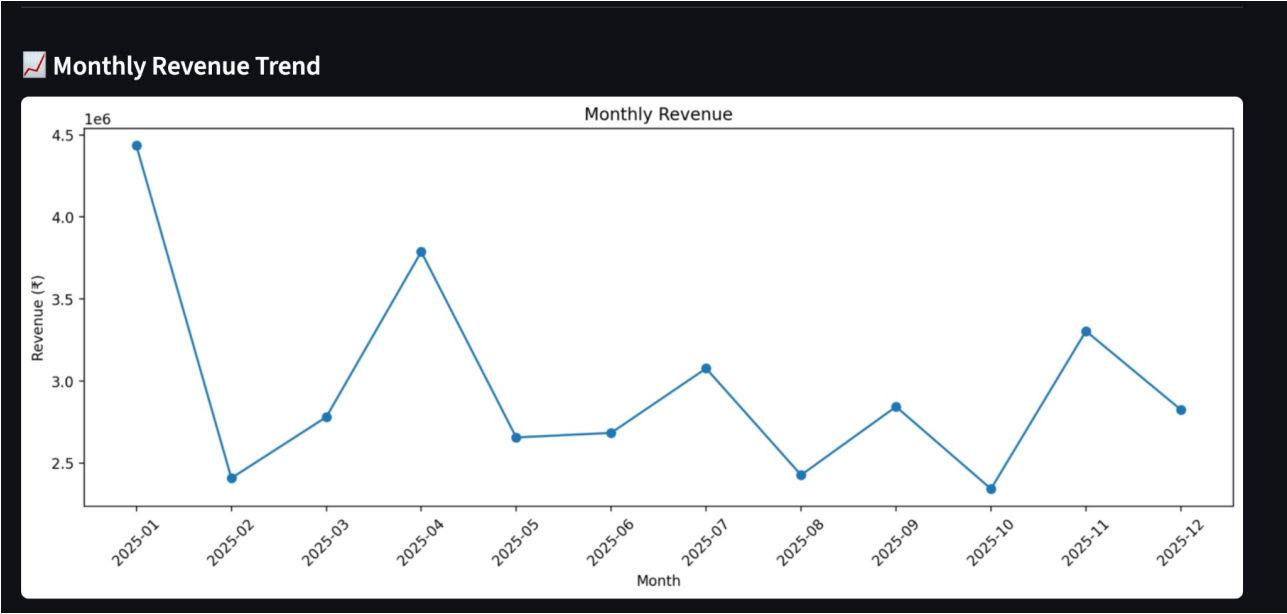


Figure 2. Monthly Revenue Trend showing revenue variation across 2025.

6.3 Category & Regional Analysis

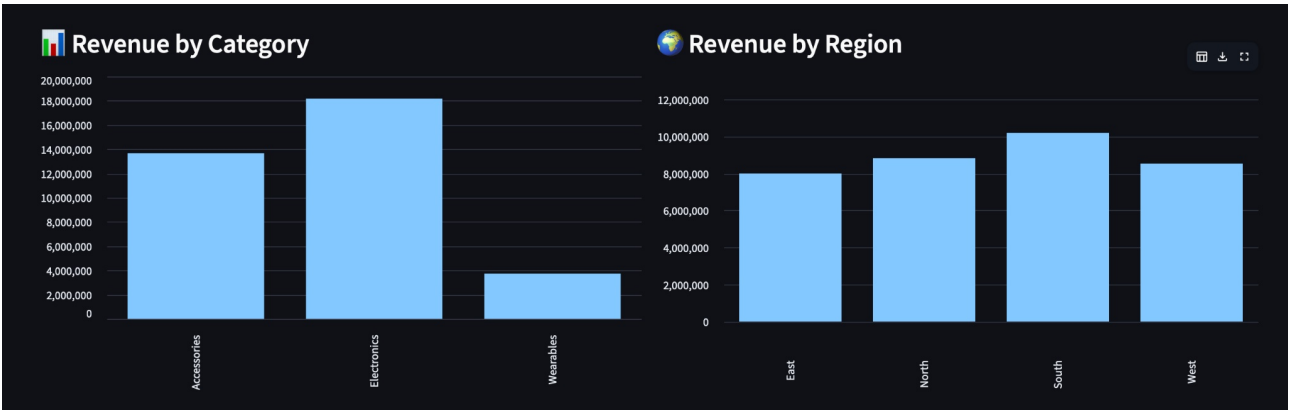


Figure 3. Revenue comparison by Category and Region.

6.4 Product Performance

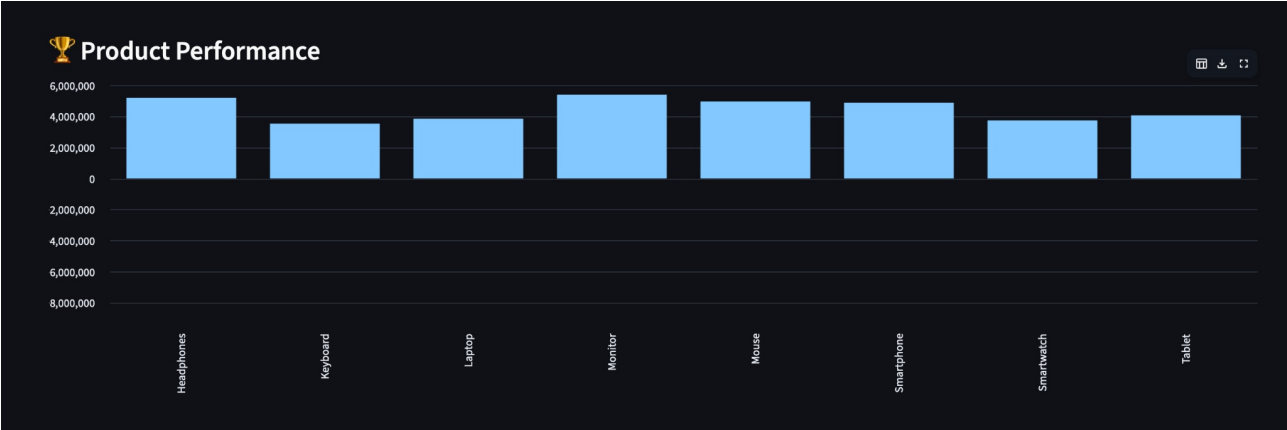


Figure 4. Product Performance comparison across the analysed products.

7. Interactive Dashboard Features

- KPI cards showing Total Revenue, Total Orders, Quantity Sold, Customers and Average Order Value.
- Monthly Revenue Trend for year-over-year monthly movement within the dataset.
- Revenue by Category comparison.
- Revenue by Region comparison.
- Product Performance ranking.
- Revenue by Payment Method.
- Top 10 Customers table with revenue, orders and quantity.
- Sidebar filters for Region, Category, Product and Payment Method.
- Expandable filtered sales-data table.
- Automatic business-insight messages highlighting leading performers.

8. Resume-Ready Project Description

E-Commerce Sales Analytics Dashboard | Python, Pandas, Matplotlib, Streamlit

- Developed an interactive e-commerce analytics dashboard analysing 1,000 orders, 198 customers and 3,051 units sold.
- Engineered KPI reporting for ■35.57M revenue and ■35,572.50 average order value using Pandas.
- Built interactive filters and automated insight generation across products, categories, regions, payment methods and monthly sales.
- Created a complete analytics workflow from CSV ingestion and transformation to visualization and Streamlit dashboard delivery.

9. Limitations & Future Enhancements

- Current analysis is descriptive and does not establish causal relationships.
- The dashboard uses a static CSV rather than a live production database.
- Profit and margin analysis would require cost-of-goods data.
- Future work can include sales forecasting, customer segmentation, anomaly detection and permanent deployment.

10. Conclusion

The project demonstrates a practical end-to-end data analytics workflow: raw transaction data is transformed into KPIs, visual analysis and an interactive business dashboard. The included screenshots provide visual evidence of the completed dashboard and its analytical outputs, making the project suitable for a GitHub portfolio and resume project section.